Online Case Study: Portfolio Management Simulation

BUS 104: Decision Analysis and Management Science

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UC RIVERSIDE | UNIVERSITY OF CALIFORNIA | School of Business
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Case Overview, Hard Purchase Cutoff & Key Dates

Putting Nonlinear Programming Into Action (10% of Course Grade)

Apply portfolio optimization to a live multiplayer simulation (Finsimco) across the **25 largest global companies** over **4 quarters** in **teams of 3 students**.

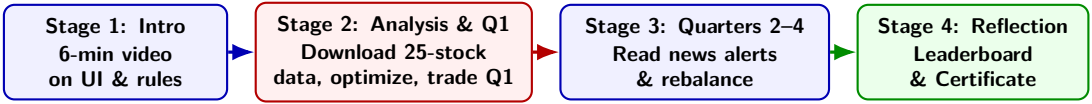
Hard Purchase Cutoff: Mon, Nov 23 (11:59 PM)

- **Course Pack:** `https://hbsp.harvard.edu/import/1456741`
- **No Exceptions:** Only students who purchase by **Mon, Nov 23 at 11:59 PM** are placed on a team. **If you miss this deadline, you cannot play and will receive 0%.**
- **Syllabus Email Rule:** You **must** register with your **exact UCR Canvas email and full name**; otherwise you will not receive credit.

Play Window & Final Deadline

- **Teams Assigned:** Teams of **3** posted on Canvas before play begins.
- **Simulation Play: Mon, Nov 30 – Fri, Dec 4** (last week of the quarter).
- **All Submissions Due: Fri, Dec 4 at 11:59 PM** (simulation completion and individual Canvas upload).

How the Simulation Works & Your Two Client Mandates



Odd Teams (1, 3, 5, . . .): Hedge Fund

- **Goal:** Beat other **Hedge Fund teams** by maximizing your portfolio **Sharpe Ratio**.
- **Tools:** May use leverage, concentrate or diversify, or hold cash.

Even Teams (2, 4, 6, . . .): Pension Fund

- **Goal:** Beat other **Pension Fund teams** by maximizing **Sharpe Ratio subject to** keeping portfolio volatility (σ_p) **below the Pension Fund median**.

Team Tip: All 3 teammates share one portfolio—designate **one member per quarter** to enter trades so you do not overwrite each other.

Syllabus Grading Scale & Peer-Adjusted Individual Scores

Syllabus Scoring Scale (10% of Grade)

Percentile	Case Score	Class Position
95–100	**100%**	Top $\sim 5\%$
90–94.9	**95%**	Next ~ 5–10%
85–89.9	**91%**	Upper tier
80–84.9	**88%**	Upper-middle
75–79.9	**86%**	Middle
>0–74.9	**80%**	**Participating Floor**
No Attempt	**0% (F)**	Non-submission / Ghost

Teams are ranked by **Sharpe Ratio** within their mandate pool (Hedge vs. Hedge; Pension vs. Pension). Valid participation guarantees a **team score** $\geq 80\%$.

Individual Submission Required to Claim Team Score

- Purchasing the case alone **does not** earn a grade.
- To claim your team's score (80%–100%), **you must individually submit** your work on Canvas by **Fri, Dec 4 (11:59 PM)**.
- **Non-Submission = 0%:** If you do not upload your individual deliverables on Canvas, you receive **0%**—even if your teammates played.

No Peer-Grading Drama

- No student can lower another student's grade.
- Your grade depends strictly on your team's objective Sharpe Ratio and **your own** Canvas submission.

Required Deliverables (Due Friday, Dec 4 at 11:59 PM)

Each Student Must Individually Submit All Items on Canvas by Fri, Dec 4 (11:59 PM)

1. **Finsimco Team Number** (entered in the Canvas submission).
2. **Personal Finsimco Certificate of Completion (PDF/Screenshot):** Downloaded from Stage 4 (*Reflection*).
3. **Optimization Model File (`.ipynb` or Excel):** Your Quarter 1 portfolio optimization file.
4. **Brief Portfolio Summary (2–4 Sentences):** Top 2 Q1 stocks and one Q2–Q4 rebalancing decision.

48-Hour No-Show Rule (Proceed Without Flaky Teammates)

- If a teammate does not respond within **48 hours** of team assignment (by **Wed, Dec 2**), **proceed without them**.
- Even 1 active student on a team can complete every stage in Finsimco, submit on Canvas, and earn the full team score ($\geq 80\%$).

Quick Python Starter: Scaling Lecture Code to 25 Stocks

In **Google Colab**, compute `mu` and `Sigma` directly from Finsimco's downloadable Excel file:

```
import pandas as pd, numpy as np
from scipy.optimize import minimize, LinearConstraint

prices = pd.read_excel("historical_data.xlsx", index_col=0)
returns = prices.pct_change().dropna()
mu = returns.mean().values * 252          # Annualized expected returns (25 stocks)
Sigma = returns.cov().values * 252        # Annualized 25x25 covariance matrix

def neg_sharpe(w, rf=0.03):               # Minimize NEGATIVE Sharpe Ratio
    return -(np.dot(mu, w) - rf) / np.sqrt(np.dot(w, np.dot(Sigma, w)))

n = len(mu)
res = minimize(neg_sharpe, x0=np.ones(n)/n, bounds=[(0, 1)]*n,
               constraints=[LinearConstraint(np.ones(n), 1.0, 1.0)])
```

Tip for Even Teams (Pension Funds): Also check your portfolio volatility $\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}$ so it stays safely below the Pension Fund median.

Summary Schedule & Action Checklist

Date & Time	Milestone	What You Must Do
Mon, Nov 23 (11:59 PM)	**HARD PURCHASE CUTOFF**	Buy course pack at `hbsp.harvard.edu/import/1456741` using your **exact Canvas email & full name**. Miss = 0%.
Before Mon, Nov 30	**Teams of 3 Assigned**	Check Canvas for your Team # (Odd = Hedge Fund, Even = Pension Fund) and message your teammates.
Mon, Nov 30 – Fri, Dec 4	**Play Window (Q1–Q4)**	Watch Intro video, optimize Q1 portfolio, and complete Q2–Q4 rebalancing trades. (48-hr No-Show Rule applies by Wed, Dec 2.)
Fri, Dec 4 (11:59 PM)	**ALL SUBMISSIONS DUE**	Submit (1) Team #, (2) Finsimco Certificate, (3) Model file, and (4) 2–4 sentence summary on Canvas.